

Victor Augusto Duraes de Faria

Charlotte, North Carolina, USA | vicadfaria@gmail.com | 919 345 8185 | [Google Scholar](#)

RESEARCH INTERESTS

Optimization models and methods, with a focus on energy systems; stochastic optimization and decision making under uncertainty; machine learning applied to optimization modeling and solution methods; capacity expansion, production cost modeling, resource adequacy, and fuel and generation portfolio management.

EDUCATION

Ph.D. in Operations Research | North Carolina State University, Raleigh, NC

2019 to July 2024 | GPA: 4.0/4.0

Dissertation: [Predictive and Prescriptive Analytics Applied to Energy Systems.](#)

M.S. in Electrical Engineering | North Carolina State University, Raleigh, NC

2022 to December 2023 | GPA: 4.0/4.0

B.S. in Electrical Engineering | Federal University of Itajuba, Brazil

2013 to 2018 | GPA: 9.0/10.0

Exchange Program in Electrical Engineering | Tennessee State University, Nashville, TN

August 2015 to August 2016 | Scholarship supported exchange | GPA: 4.0/4.0

RESEARCH AND PROFESSIONAL EXPERIENCE

Senior Data Science Consultant | Duke Energy, USA

August 2025 to present | Supervisor: [Patrick O'Connor](#)

- Develop ORBIT (Overnight Resource Balancing & Intelligent Trading), an hourly economic dispatch optimization tool that updates operating schedules as load forecasts, outages, power and gas trades, and fuel constraints change.
- Develop Dynamic Dispatch methods to manage fuel inventory positions through fuel cost adjustments that account for changing fuel prices, delivery logistics, transportation limits, and interactions among generation technologies.
- Enhance unit commitment and production cost models to evaluate dispatch strategies, reliability, and operating costs across thermal generation, renewable generation, and pumped storage.

Researcher III, Model Engineering | National Renewable Energy Laboratory (NREL; now National Laboratory of the Rockies), USA

August 2024 to August 2025 | Supervisors: [Ilya Chernyakhovskiy](#), [Brian Sergi](#), [Patrick Brown](#), [Paul Denholm](#), and [Trieu Mai](#)

- Developed uncertainty quantification workflows for the ReEDS capacity expansion model to analyze technology cost and fuel price uncertainty. The workflows are included in the public ReEDS model repository.
- Provided technical analysis for the West Virginia Office of Energy on grid modernization, generation portfolio transition, and reliability.
- Modeled generation expansion scenarios under rapid load growth for PJM reliability and planning studies.
- Conducted resource adequacy assessments using probabilistic methods to evaluate reliability across future generation portfolios.
- Integrated and analyzed generation and demand datasets to support power system modeling and scenario analysis.

Research Assistant | North Carolina State University, USA

Department of Civil, Construction, and Environmental Engineering | August 2019 to July 2024 | Supervisors: [Anderson Rodrigo de Queiroz](#) (2019 to 2024) and [Joseph DeCarolis](#) (2019 to 2023)

- Developed optimization and statistical models in Python, MATLAB, and R for the economic assessment of energy technologies.
- Modeled the financial and operational consequences of hurricanes and other extreme events for energy infrastructure.
- Developed a stochastic capacity expansion framework incorporating hurricane damage and infrastructure vulnerability into electricity planning.
- Evaluated capacity expansion and renewable energy integration strategies for the North Carolina electricity system.
- Developed mean variance and conditional value at risk portfolio models to evaluate generation siting and investment strategies.
- Applied generative adversarial networks to renewable generation scenario development for stochastic portfolio optimization.
- Developed routing optimization models to assess transportation network vulnerability during flooding.

MC&E, Marangon Consulting & Engineering | Brazil

2017 to 2019 | Advisors: [Anderson Rodrigo de Queiroz](#), [Jose Marangon Lima](#), and [Luana Marangon Lima](#)

- Applied machine learning methods in Python and TensorFlow to electrical system planning and analysis.
- Developed neural network models and automated workflows for streamflow forecasting at Brazilian hydropower plants.

EXCEN, Centre of Excellence on Energy Efficiency | Brazil

March 2017 to June 2018 | Advisor: [Edson da Costa Bortoni](#)

- Developed mathematical models of synchronous machine behavior during sudden short circuit tests.
- Validated model behavior through experimental testing on synchronous machines.

Department of Industrial and Operations Engineering | University of Michigan, USA

June 2016 to August 2016 | Advisor: [Pascal Van Hentenryck](#)

- Investigated convex relaxation approaches for optimal power flow and evaluated their computational performance.

GESis, Group of Systems Engineering | Brazil

2014 to 2017 | Advisors: [Anderson Rodrigo de Queiroz](#), [Jose Marangon Lima](#), and [Luana Marangon Lima](#)

- Developed C++ models for analysis of the Brazilian electrical system.
- Evaluated optimization formulations for allocating firm energy rights among hydropower producers.
- Developed cooperative game theory approaches for firm energy allocation.
- Analyzed the effects of changing hydrological conditions on hydropower revenues.

PUBLICATIONS

Google Scholar: 446 citations (September 2026). † Student mentored by V.A.D. de Faria.

Journal Articles

- [J1.](#) Y.H. Chou†, **V.A.D. de Faria**, A.R. de Queiroz, (2026). [Transfer learning and convolutional neural networks for solar photovoltaic systems detection in distributed power grids](#), Renewable Energy.
- [J2.](#) **V.A.D. de Faria**, A.R. de Queiroz, A. Sankarasubramanian, R. Ranjithan, (2026). [Routing Optimization Framework for Exploring Time-Varying Urban Road Network Vulnerability under Floods](#), Journal of Infrastructure Systems.

- J3. M. Maceda, R. Miller†, **V.A.D. de Faria**, M. Bryant, C. Vermillion, A.R. de Queiroz, (2026). [Fused Portfolio Optimization for Harnessing Marine Renewable Energy Resources](#), Energy.
- J4. N. Jamaledin, **V.A.D. de Faria**, A.R. de Queiroz, M. Gabr, (2025). [Optimizing co-located offshore wave and wind energy systems: Feasibility, design, and portfolios assessment](#), Journal of Cleaner Production.
- J5. M. Nicoli, **V.A.D. de Faria**, A.R. de Queiroz, L. Savoldi, (2024). [Modeling energy storage in long-term capacity expansion energy planning: an analysis of the Italian system](#), Journal of Energy Storage.
- J6. **V.A.D. de Faria**, N. Jamaledin, A.R. de Queiroz, M. Gabr, (2024). [Bayesian modeling and mechanical simulations for fragility curve estimation of the mooring system of marine hydrokinetic devices](#), Applied Ocean Research.
- J7. **V.A.D. de Faria**, A.R. de Queiroz, J.F. DeCarolis, (2023). [Scenario Generation and Risk-averse Stochastic Portfolio Optimization Applied to Offshore Renewable Energy Technologies](#), Energy.
- J8. G. Aquila, L.B.S. Morais, **V.A.D. de Faria**, J.W.M. Lima, L.M.M. Lima, A.R. de Queiroz, (2023). [An Overview of Short-Term Load Forecasting for Electricity Systems Operational Planning: Machine Learning Methods and the Brazilian Experience](#), Energies.
- J9. L.B.S. Morais, G. Aquila, **V.A.D. de Faria**, L.M.M. Lima, J.W.M. Lima, A.R. de Queiroz, (2023). [Short-term load forecasting using neural networks and global climate models: An application to a large-scale electrical power system](#), Applied Energy.
- J10. **V.A.D. de Faria**, A.R. de Queiroz, J.F. DeCarolis, (2022). [Optimizing Offshore Renewable Portfolios Under Resource Variability](#), Applied Energy.
- J11. **V.A.D. de Faria**, A.R. de Queiroz, L.M.M. Lima, J.W.M. Lima, B.C. Silva, (2022). [An assessment of multi-layer perceptron networks for streamflow forecasting in large-scale interconnected hydrosystems](#), International Journal of Environmental Science and Technology.
- J12. **V.A.D. de Faria**, J.V. Bernardes, E.C. Bortoni, (2020). [Parameter estimation of synchronous machines considering field voltage variation during the sudden short-circuit test](#), International Journal of Electrical Power & Energy Systems.
- J13. A.R. de Queiroz, **V.A.D. de Faria**, L.M.M. Lima, J.W.M. Lima, (2019). [Hydropower Revenues under the Threat of Climate Change in Brazil](#), Renewable Energy.
- J14. E.C. Bortoni, J.V. Bernardes, P.V.V. da Silva, **V.A.D. de Faria**, P.A.V. Vieira, (2019). [Evaluation of manufacturers strategies to obtain high-efficient induction motors](#), Sustainable Energy Technologies and Assessments.
- J15. **V.A.D. de Faria**, A.R. de Queiroz, L.M.M. Lima, J.W.M. Lima, (2018). [Cooperative Game Theory and Last Addition Method in the Allocation of Firm Energy Rights](#), Applied Energy.

Technical Reports

- R1. C. Avraam, A. Ayad, P. Brown, V. Carag, Y. Chen, I. Chernyakhovskiy, S. Cohen, W. Cole, S. Dhulipala, **V.A.D. de Faria**, et al., (2025). [Regional Energy Deployment System \(ReEDS\) Model Documentation: 2025](#), National Renewable Energy Laboratory (NREL).
- R2. P. Denholm, **V.A.D. de Faria**, J. Frost, (2024). [How the U.S. Power Grid Kept the Lights on in Summer 2024](#), National Renewable Energy Laboratory (NREL).

Conference Papers

- C1. **V.A.D. de Faria**, B. Sergi, P.R. Brown, (2026). Multidimensional uncertainty assessment in a capacity expansion model for the U.S. power system, 19th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), Salt Lake City, UT, USA (accepted, in editorial production IEEE). Funded by the U.S. Department of Energy (DOE) Grid Deployment Office.

Work Under Review

- W1. **V.A.D. de Faria**, J. Jorgenson, G. Brinkman, T. Mai, M. Vanatta, R. Fuchs (2026). Planning for Resource Adequacy Under Rising VRE Levels: A Cost-Driven Approach. National Renewable Energy Laboratory. Under DOE internal review.

PRESENTATIONS AND POSTERS

** Presenter.*

- P1. **V.A.D. de Faria**, B. Sergi, P.R. Brown* (2026). Multidimensional uncertainty assessment in a capacity expansion model for the U.S. power system. Oral presentation, 19th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS), Salt Lake City, UT, September 22, 2026.
- P2. N. Jamaledin*, **V.A.D. de Faria**, A.R. de Queiroz, M. Gabr (2024). Exploring the feasibility of co-located marine renewable energy systems: A mooring system and economic analysis in the North Carolina coast. Poster, UMERG+METS 2024 Conference, Duluth, MN, August 7, 2024.
- P3. **V.A.D. de Faria***, A.R. de Queiroz (2023). Energy systems capacity expansion planning under the risk of hurricane damage. Oral presentation, INFORMS Annual Meeting, Phoenix, AZ, October 15 to 18, 2023.
- P4. M. Nicoli*, **V.A.D. de Faria**, A.R. de Queiroz, L. Savoldi (2023). Modeling energy storage in capacity expansion models: An application to TEMA Italy. Presentation, INFORMS Annual Meeting, Phoenix, AZ, October 15 to 18, 2023.
- P5. **V.A.D. de Faria***, N. Jamaledin, A.R. de Queiroz, M. Gabr, M. Muglia (2022). Assessing the risk of hurricane damage to marine hydrokinetic devices. Poster, UMERG+METS 2022 Conference, Portland, OR, September 14, 2022.

RESEARCH FUNDING

Sponsored projects on which V.A.D. de Faria served as a researcher. Awards were made to the principal investigators (PI) listed.

- F1. U.S. Department of Energy, Grid Deployment Office, through NREL. Uncertainty characterization in the ReEDS capacity expansion model. 2024 to 2026. Role: model developer and first author of the resulting paper (C1).
- F2. National Science Foundation, Convergence Accelerator Phase II, Award #2033607. The Urban Flooding Open Knowledge Network (UF-OKN): Delivering Flood Information to Anyone, Anytime, Anywhere. Lead institution: University of Cincinnati (PI: L. Yeghiazarian); NC State Co-PI: S. Arumugam. \$5,299,998 total multi-institution award, 2020 to 2024. Role: graduate researcher, flood-informed routing optimization and risk assessment ([J2](#)).
- F3. North Carolina Renewable Ocean Energy Program (NCROEP), FY2023 to 2024. Fused Portfolio Optimization for Harnessing Marine Renewable Energy Resources: Exploring Opportunities for Improving Energy Density and Levelized Cost of Electricity. PI: A.R. de Queiroz. \$50,000. Role: graduate researcher, student mentor, and contributor to proposal writing ([J3](#)).
- F4. NCROEP, FY2023 to 2024. Exploring the Feasibility of Co-Located Marine Renewable Energy Systems: A Mooring System and Economic Analysis in the North Carolina Coast. PI: A.R. de Queiroz. \$50,000. Role: graduate researcher and contributor to proposal writing ([J4](#)).
- F5. NCROEP, FY2022 to 2023. Fused Portfolio, Site, and Device Sizing Optimization for Harnessing North Carolina's Coastal Renewable Energy Resources. PI: C. Vermillion. \$59,000 (NC State \$48,836; NCCU \$10,164). Role: graduate researcher and contributor to proposal writing ([J3](#)).
- F6. NCROEP, FY2022 to 2023. Mooring System Analysis and Fragility Curve Estimation: The Economic Impact of Low Probability High Impact Events on Ocean Current Devices. PI: M.A. Gabr. \$52,923. Role: graduate researcher and contributor to proposal writing ([J6](#)).

- F7. NCROEP, FY2021 to 2022. Assessing the Risk of Hurricane Damage to Marine Hydrokinetic Devices. PI: J.F. DeCarolís. \$59,728. Role: graduate researcher and contributor to proposal writing ([J6](#)).
- F8. NCROEP, FY2020 to 2021. Analyzing Investments in NC Offshore Renewable Energy Under Uncertain Resource Availability and Hurricane Damages. PI: J.F. DeCarolís; Co-PI: A.R. de Queiroz. \$38,236. Role: graduate research assistant named in the proposal ([J7](#)).
- F9. NCROEP, FY2019 to 2020. Optimizing Investments in NC Offshore Renewable Energy. PI: J.F. DeCarolís; Co-PI: A.R. de Queiroz. \$32,420. Role: graduate research assistant ([J10](#)).

RESEARCH MENTORING

- M1. [Rob Miller](#), Ph.D. student, 2023 to 2024. Marine renewable energy portfolio, site, and device sizing optimization under the NCROEP ocean energy projects. Co-author of [J3](#).
- M2. [Yen-Hsi Chou](#), Ph.D. student, 2023 to 2025. Transfer learning and convolutional neural networks for detecting solar photovoltaic systems in distributed power grids. First author of [J1](#).

PROFESSIONAL SERVICE

Journal reviewer: More than 40 manuscript reviews since 2021. Journals include Joule; Applied Energy; IEEE Transactions on Energy Conversion; Ocean Engineering; IET Generation, Transmission & Distribution; Electronics; The Journal of Supercomputing; Energy Engineering; Journal of the Association of Energy Engineers; Forecasting; and Frontiers in Energy Research.

RESEARCH PROJECTS

ORBIT Production Cost Model

August 2025 to present | Duke Energy | Supervisor: [Patrick O'Connor](#)

Develop ORBIT (Overnight Resource Balancing & Intelligent Trading), a production cost optimization framework linking generation scheduling with natural gas constraints and power trading decisions. Incorporate Operational Flow Orders (OFOs) and changing gas availability to evaluate their implications for generation flexibility and operating costs. Investigate coordinated adjustments to gas nominations, generation dispatch, and power trades as market prices, load forecasts, and outages evolve, supporting operational decisions across interconnected gas and electricity systems.

Dynamic Dispatch Model for Duke Energy Operations

August 2025 to present | Duke Energy | Supervisor: [Patrick O'Connor](#)

Develop optimization methods that translate fuel inventory positions, market prices, and delivery constraints into dynamic fuel cost signals for generation dispatch. Investigate how these adjustments influence fuel consumption, inventory trajectories, and generation costs. Integrate fuel delivery schedules, transportation limitations, and interactions among generation technologies to evaluate the tradeoffs between immediate dispatch economics and future fuel availability, supporting coordinated inventory management and generation planning.

ReEDS Enhancements to Characterize Uncertainty

August 2024 to August 2025 | NREL | Supervisors: [Brian Sergi](#) and [Patrick Brown](#)

Funded by the U.S. Department of Energy Grid Deployment Office, 2024 to 2025

Developed Monte Carlo simulation workflows in GAMS and Python to characterize uncertainty in ReEDS capacity expansion results. Integrated these workflows with the model to evaluate how uncertain technology costs and fuel prices affect generation investment decisions and planning outcomes. Related publication: C1.

Resource Adequacy Under Rising Variable Renewable Energy Levels

October 2024 to July 2025 | NREL | Supervisor: [Trieu Mai](#)

Contributed to an iterative modeling workflow linking ReEDS capacity expansion decisions with PRAS probabilistic resource adequacy assessments to evaluate cost and reliability across future power systems. Used the coupled framework to assess how capacity investments can address reliability needs as variable renewable generation increases. Related manuscript: [W1](#).

Capacity Expansion Planning Under Hurricane Risk in North Carolina

January 2023 to July 2024 | North Carolina State University | Supervisor: [Anderson Rodrigo de Queiroz](#)

Developed a multistage stochastic extension of TEMOA in Python to evaluate generation investment under hurricane risk in North Carolina. Integrated FEMA HAZUS data and engineering fragility curves to represent damage across electricity generation technologies, enabling capacity expansion decisions to account for infrastructure vulnerability alongside cost.

Joint Portfolio, Site, and Device Sizing Optimization for Marine Energy

August 2022 to August 2023 | Supervisors: [Anderson Rodrigo de Queiroz](#) and [Chris Vermillion](#)

Funded by the North Carolina Renewable Ocean Energy Program, 2022 to 2023

Developed integrated optimization models in Python and MATLAB to jointly select marine renewable energy technologies, sites, and device sizes. Formulated the models to identify cost minimizing portfolios, connecting engineering design choices with resource location and investment decisions. Related publication: [J3](#).

Mooring System Fragility and Economic Risk for Ocean Current Devices

August 2021 to August 2023 | Supervisors: [Anderson Rodrigo de Queiroz](#) and [Mo Gabr](#)

Funded by the North Carolina Renewable Ocean Energy Program, 2021 to 2023

Combined Bayesian modeling of oceanographic data with mechanical simulations to estimate mooring system fragility curves for ocean current devices. Quantified failure probabilities under extreme ocean conditions, providing inputs for resilient mooring design and the assessment of financial risk. Related publication: [J6](#).

Urban Flooding Open Knowledge Network

August 2020 to August 2023 | Supervisors: [Anderson de Queiroz](#), [Ranji Ranjithan](#), and [Sankar Arumugam](#)

Funded by the National Science Foundation Convergence Accelerator (Award #2033607)

Developed methods to translate hydrological forecasts into road closures and speed reductions, and implemented routing optimization models that account for evolving flood conditions. The models assess road network vulnerability and accessible routes, supporting analysis of transportation disruptions for emergency response and logistics planning. Related publication: [J2](#).

Offshore Renewable Energy Investment Under Resource and Hurricane Uncertainty

August 2020 to August 2021 | Supervisors: [Anderson Rodrigo de Queiroz](#) and [Joseph DeCarolis](#)

Funded by the North Carolina Renewable Ocean Energy Program, 2020

Developed generative adversarial network models to produce thousands of annual wind, wave, and ocean current generation scenarios that capture resource variability and correlations. Integrated these scenarios with risk averse stochastic portfolio optimization to evaluate investment locations and financial risk under uncertain electricity generation. Related publication: [J7](#).

Offshore Renewable Portfolio Optimization for North Carolina

August 2019 to August 2020 | Supervisors: [Anderson Rodrigo de Queiroz](#) and [Joseph DeCarolis](#)

Funded by the North Carolina Renewable Ocean Energy Program, 2019

Adapted mean variance portfolio optimization to represent offshore energy technologies and sites as investment assets. Identified portfolios that minimize energy output variability at a given cost, making cost and output stability explicit criteria for offshore energy planning. Related publication: [J10](#).

Neural Network Streamflow Forecasting for Hydropower Plants

2018 to 2019 | Supervisors: [Anderson Rodrigo de Queiroz](#), [Jose Marangon Lima](#), and [Luana Marangon Lima](#)

Developed and trained neural network models to forecast streamflow for more than 140 hydropower plants in Brazil, supporting daily assessment of water availability. Automated data ingestion, daily model execution, and performance reporting using data from Brazil's National Electric System Operator (ONS) and the U.S. National Oceanic and Atmospheric Administration (NOAA). Related publication: [J11](#).

Field Voltage Variation and Synchronous Machine Parameter Estimation

2018 to 2019 | Supervisor: [Edson da Costa Bortoni](#)

Bachelor's thesis, 2019

Modeled and quantified how field winding voltage variation during sudden short circuit tests affects synchronous machine parameter estimation. Identified conditions under which the constant voltage assumption introduces estimation errors, establishing limits for its use in interpreting test results. Related publication: [J12](#).

Cooperative Game Theory for Firm Energy Rights Allocation

2016 to 2017 | Supervisors: [Anderson Rodrigo de Queiroz](#), [Jose Marangon Lima](#), and [Luana Marangon Lima](#)

Developed and implemented optimization formulations based on cooperative game theory for allocating firm energy rights in Brazil. Demonstrated computational feasibility at the scale of the national hydropower system and compared allocation outcomes, providing a framework for evaluating how energy rights are distributed among producers. Related publication: [J15](#).

Climate Change and Hydropower Revenues in Brazil

2015 to 2017 | Supervisors: [Anderson Rodrigo de Queiroz](#), [Jose Marangon Lima](#), and [Luana Marangon Lima](#)

Developed energy system optimization models in C++ to simulate the national hydropower system under alternative climate scenarios. Quantified how changes in water availability affect electricity costs and hydropower producer revenues, connecting future hydrological conditions with financial exposure in Brazil. Related publication: [J13](#).

Convex Relaxation of Optimal Power Flow

May 2016 to August 2016 | University of Michigan | Supervisor: Pascal Van Hentenryck

Modeled convex relaxation approaches for optimal power flow and benchmarked their computational performance, assessing how formulation choices affect the computational effort required to solve power system operating problems.

TECHNICAL SKILLS AND LANGUAGES

Optimization: Linear, nonlinear, integer, and stochastic programming; portfolio optimization; capacity expansion and production cost modeling.

Statistical and machine learning methods: Bayesian inference and MCMC; neural networks; computer vision; time series forecasting; detection and estimation.

Programming and modeling: Python, MATLAB, R, C++, AMPL, and GAMS. Libraries and frameworks: Pyomo, PyTorch, and TensorFlow. Optimization software: Gurobi, CPLEX, and COIN OR.

Additional software: ArcGIS, SolidWorks, and JMP.

Languages: Portuguese, native; English, fluent.